

Computational perspectives on the Classical Decision Problem

Lecture 7: The Two-Variable Fragment with Counting

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- The **two-variable fragment with counting** ($\mathcal{C}^2$) is the set of function-free first-order formulas featuring only the variables x and y , but with counting quantifiers $\exists_{\leq M}$, $\exists_{\geq M}$, $\exists_{=M}$ allowed.

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- Example formula:

No artist envies more than 13 beekeepers that despise exactly 647 carpenters

$$\forall x(\text{artist}(x) \rightarrow \exists_{\leq 13} y(\text{beekeeper}(y) \wedge \exists_{=647} x(\text{carpenter}(x) \wedge \text{despise}(x, y)) \wedge \text{envy}(x, y))).$$

- Equality is definable in $\mathcal{C}^2$:

$$\forall x(r(x, x) \wedge \exists_{\leq 1} y \, r(x, y)).$$

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- Individual constants do not add useful expressive power.

$$\exists x \, r(x, c) \implies \exists_{=1} x \, p_c(x) \wedge \exists x \exists y (r(x, y) \wedge p_c(y)).$$

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- The truth-values of 0-ary predicates can be guessed, so we **disallow** them.
- Henceforth, signatures will be assumed to consist only of predicates of arity 1 or 2, but $=$ may be present.

- Let r be a binary relation ('successor').
- We can state in $\mathcal{C}^2$ that every object x has a successor, some object has no predecessor, and no object has more than one predecessor.

$$\forall x \exists y \ r(x, y) \wedge \exists x \forall y \neg r(y, x) \wedge \forall x \exists \leq 1 y \ r(y, x).$$

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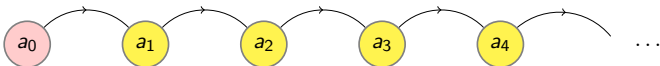
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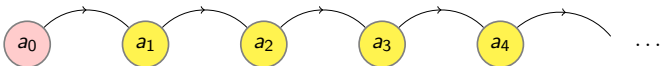
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- On the other hand, the above structure is a model.
- Thus, $\mathcal{C}^2$ lacks the **finite model property** (FMP).

- As a further example, let $X_{n-1}, \dots, X_0$ be unary predicates encoding digits in an n -digit number.

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- In any structure interpreting these predicates each object is associated with a number d in the range $0 \leq d < 2^n - 1$; call this number its '**depth**'.
- The following formulas says that the depth of y is one greater than the depth x :

$$\bigwedge_{i=0}^{n-1} \left(\left(\bigvee_{j=0}^{i-1} \neg X_j(x) \right) \leftrightarrow (X_i(x) \leftrightarrow X_i(y)) \right).$$

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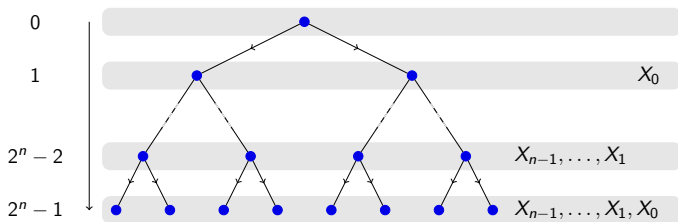
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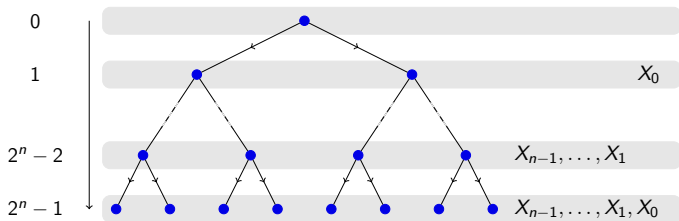
- There is an x with depth 0, and everything with depth $< 2^n - 1$ has at least two successors, and nothing has more than one predecessor:

$$\begin{aligned} \exists x \bigwedge_{i=0}^n \neg X_i(x) \wedge \\ \forall x \left(\left(\bigvee_{i=0}^{n-1} \neg X_i(x) \rightarrow \exists_{[\geq 2]} y \, r(x, y) \right) \wedge \right. \\ \left. \forall x \exists_{[\leq 1]} y \, r(y, x) \right). \end{aligned}$$

- Any model of the above formulas embeds the structure

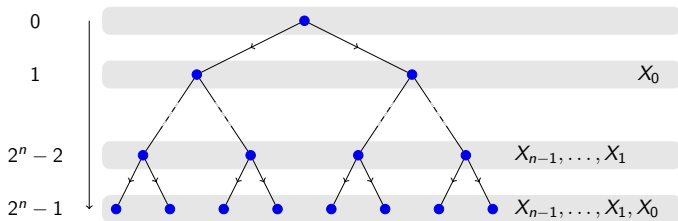


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- On the other hand, the above structure is a model.
- Thus, finitely satisfiable formulas of $\mathcal{C}^2$ can force 'doubly-exponential' models (Grädel, Otto and Rosen, 1997).

- If φ is a sentence of $\mathcal{C}^2$, then we can compute, in polynomial time, a sentence ψ of the form

$$\forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{1 \leq h \leq m} \forall x \exists_{=M_h} y (\beta_h(x, y) \wedge x \neq y),$$

where α and the β_h are quantifier-free, such that φ and ψ are satisfiable over the same domains of size $> M = \max_{h=1}^m M_h$.

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- The proof is essentially the same as for $\mathcal{FO}^2$.
- We refer to such formulas as ‘normal-form’ formulas.

- Recall the apparatus of atomic *k*-types introduced in Lecture 3.
- Let Σ be a finite, purely relational signature in which all predicates have arity at least 1.
 - A *k*-type (over Σ) is a maximal consistent set of equality-free literals over Σ involving no variables other than $x_1, \dots, x_k$.
- $\mathfrak{A}$ is a structure interpreting Σ , and $\bar{a} = a_1, \dots, a_k$ are elements of A , then $\text{tp}^{\mathfrak{A}}[\bar{a}]$ denotes the unique *k*-type $\tau(x_1, \dots, x_k)$ such that $\mathfrak{A} \models \tau[a_1, \dots, a_k]$.

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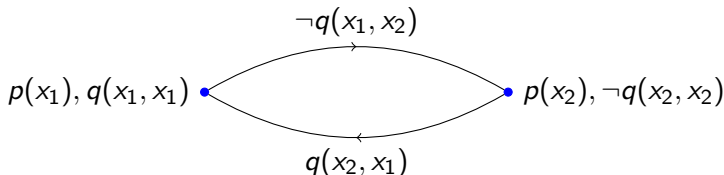
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- The following is a 2-type:

$$\tau := \{p(x_1), p(x_2), q(x_1, x_1), \neg q(x_1, x_2), q(x_2, x_1), \neg q(x_2, x_2)\}.$$

We might picture τ as follows:



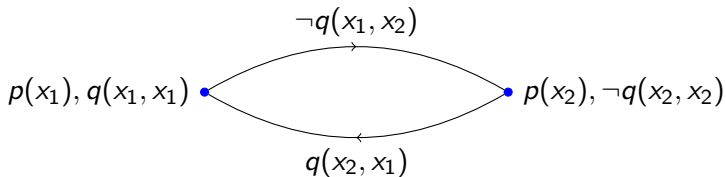
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We might picture τ as follows:



- Observe that, in this case $\models \tau \rightarrow \pi$.

- If $|\Sigma| = s$ predicates, then:
 - the number of 1-types over Σ is exactly 2^s .
 - the number of 2-types over Σ is at most 2^{4s} .
- In that case, we can enumerate the 1-types over Σ as

$$\pi_1, \dots, \pi_I;$$

and similarly for the 2-types.

- Now suppose we fix a normal-form-formula

$$\varphi := \forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{1 \leq h \leq m} \forall x \exists y (=_{M_h}) (\beta_h(x, y) \wedge x \neq y),$$

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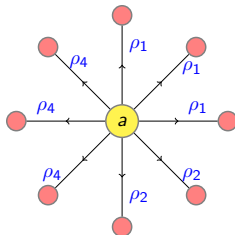
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- If neither τ nor τ^{-1} is a ray-type, we call τ **dark**.
- Evidently, we can enumerate the ray-types as, say,

$$\underbrace{\rho_1, \dots, \rho_{J^*}}_{\text{invertible}}, \underbrace{\rho_{J^*+1}, \dots, \rho_J}_{\text{non-invertible}}.$$

- If $\mathfrak{A}$ is a model of

$$\varphi := \forall x \forall y (\alpha \vee x = y) \wedge \bigwedge_{1 \leq h \leq m} \forall x \exists_{=M_h} y (\beta_h(x, y) \wedge x \neq y),$$

the number of ray-types 'emitted' by any $a \in A$ is bounded:



- Specifically,

$$|\{b \in A \setminus \{a\} : \mathfrak{A} \models \rho_j[a, b]\}| \leq M = \max_{h=1}^m M_h.$$

- Henceforth, we fix M , β_h and other components of φ .

- Thus, if $\mathfrak{A}$, then any element $a \in A$ can be characterized by a data-structure

$$\sigma = \langle \pi, v_1, \dots, v_J \rangle$$

where $\pi = \text{tp}^{\mathfrak{A}}[a]$ and

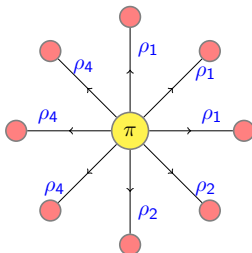
$$v_j = |\{b \in A \setminus \{a\} : \text{tp}^{\mathfrak{A}}[a, b] = \rho_j\}|.$$

Notice that $v_j > 0$ implies $\models \rho_j \rightarrow \pi$.

- More abstractly, say that an **star-type** over Σ (with respect to φ) is any structure

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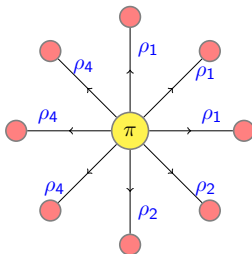
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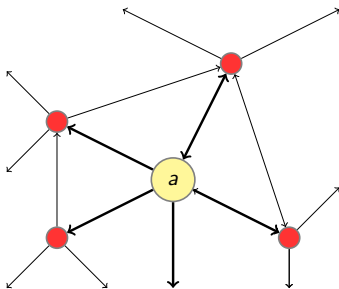
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- There is a bounded number of such star-types, so they can be listed as, say

$$\sigma_1, \dots, \sigma_K.$$

- We can think of a model $\mathfrak{A} \models \varphi$ as a jigsaw puzzle made by fitting together copies of star-types:



- Hence $\mathfrak{A}$ can be characterized (not up to isomorphism) *statistically*, by specifying how often each of the $\sigma_1, \dots, \sigma_N$ is realized.

- Fix the enumeration of star-types (for φ , over Σ)

$$\sigma_1, \dots, \sigma_K.$$

Any **finite** model $\mathfrak{A}$ of φ realizing exactly the star-types $\sigma_1, \dots, \sigma_K$ can be characterized by a vector of positive integers

$$(w_1, \dots, w_K),$$

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- We can think of this vector as the **histogram** of $\mathfrak{A}$.
- Question: when is a given vector $(w_1, \dots, w_K)$ of natural numbers the histogram of some structure?
- Answer: when it satisfies a certain set of **linear equations**.

- To explain what these equations look like, consider the k th star-type ($1 \leq k \leq K$):

$$\sigma_k = \langle \pi, v_1, \dots, v_j, \dots, v_J \rangle.$$

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- That is:

$$\sum_{k=1}^K c_{j,k} w_k = \sum_{k=1}^K c_{j',k} w_k.$$

- Here is another condition. Recall the enumeration of 1-types π_i ($1 \leq i \leq I$). Define

$$a_{i,k'} = \begin{cases} 1 & \text{if } \sigma_{k'} = \langle \pi_1, \dots \rangle \\ 0 & \text{otherwise.} \end{cases}$$

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- Then the total number of elements of $\mathfrak{A}$ with 1-type π_i is $\sum_{k'=1}^K a_{i,k'} w_{k'}$.
- Now suppose that σ_k is a star-type that emits a total of s rays to elements having 1-type π . Then the histogram of $\mathfrak{A}$ satisfies

$$w_k > 0 \implies \sum_{k'=1}^K a_{i,k'} w_{k'} \geq s.$$

That is: if there are any elements of star-type σ_k , there have to be enough elements of 1-type π_i to absorb the rays they emit.

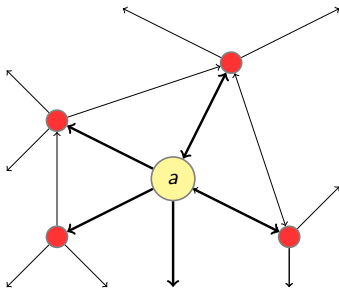
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- Given φ (in normal form), we can write down a system $\mathcal{Q}(w_1, \dots, w_K)$ of linear Diophantine clauses in such that, if φ has a finite model, the histogram of $\mathfrak{A}$ is a solution of $\mathcal{Q}$.

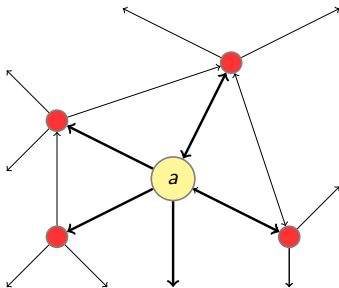
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- We would like to prove a converse: if $(w_1, \dots, w_K)$ is a solution of $\mathcal{Q}$, then it is the histogram of a finite model $\mathfrak{A} \models \varphi$.

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- In this way the problem $\text{FinSat}(\mathcal{C}^2)$ is reduced to the problem of determining the feasibility of a system of linear Diophantine clauses.

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- We thus obtain an exponential-sized integer linear programming problem.
- Hence $\text{FinSat}(\mathcal{C}^2)$ is in NEXP TIME .
- Also, we obtain a small-model property: if a $\mathcal{C}^2$ -formula is finitely satisfiable, it has a model of at most doubly exponential size.

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 - if $\mathcal{E}$ has a solution over $\mathbb{N}^*$, then it has a solution in which only exponentially many values are non-zero;
 - if $\mathcal{E}$ has a solution over $\mathbb{N}^*$, it has a solution in which the number of bits is polynomially bounded in the size of $\mathcal{E}$.
- Hence $\text{Sat}(\mathcal{C}^2)$ is in NEXP TIME .

- Putting everything together:

Theorem

The problems $\text{Sat}(\mathcal{C}^2)$ and $\text{FinSat}(\mathcal{C}^2)$ are both NEXPTIME-complete.

- History of $\mathcal{C}^2$.
 - $\text{Sat-}\mathcal{C}^2$ and $\text{Fin-Sat-}\mathcal{C}^2$ are decidable (Grädel, Otto and Rosen 1997).
 - $\text{Sat-}\mathcal{C}^{2,p}$ is in NEXPTIME . (Pacholski, Tendera and Szwoast 1997, 1999). Note: p is a fixed bound on the numerical quantifier subscripts.
 - $\text{Sat-}\mathcal{C}^2$ and $\text{Fin-Sat-}\mathcal{C}^2$ are both in NEXPTIME (IEP-H, 2005). The same method works uniformly for satisfiability and finite satisfiability.
 - The latter proof does not employ the Carathéodory-type lemma to reduce the complexity—it uses a difficult model-surgery process to reduce the numbers of star-types occurring in models.

- Recall that, in the **guarded fragment**, $\mathcal{GF}$, quantification is limited to the forms

$$\forall x(p(x, \bar{y}) \rightarrow \psi)$$

$$\exists x(p(x, \bar{y}) \wedge \psi)$$

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- The **guarded two-variable fragment with counting**, $\mathcal{GC}^2$ is the fragment of $\mathcal{C}^2$ with:
 - no individual constants
 - only **guarded** (counting) quantification

$$\begin{aligned}\forall u \varphi \quad & \varphi \text{ has at most one free variable} \\ \exists u \varphi \quad & \varphi \text{ has at most one free variable} \\ \forall u(\alpha \rightarrow \psi) \\ Qu(\alpha \wedge \psi),\end{aligned}$$

where α is an atom s.t. $\text{Vars}(\alpha) = \{x, y\}$ and Q is one of $\exists, \exists_{\leq C}, \exists_{=C}, \exists_{\geq C}$.

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- For example:

$$\forall x(\text{artist}(x) \rightarrow \exists_{\leq 17} y(\text{despise}(x, y) \wedge \text{beekeeper}(y))).$$

- Properties of $\mathcal{G}$ and $\mathcal{G}^k$ ([Grädel 1999](#)):
 - $\mathcal{GF}$ has the finite model property;
 - $\text{Sat-}\mathcal{GF}$ ($= \text{Fin-Sat-}\mathcal{GF}$) is 2-EXPTIME-complete;
 - $\text{Sat-}\mathcal{GF}^k$ ($= \text{Fin-Sat-}\mathcal{GF}^k$) is EXPTIME-complete.
- Properties of $\mathcal{GC}^2$
 - FMP fails;
 - finite models can be doubly exponentially large;
 - $\text{Sat-}\mathcal{GC}^2$ is EXPTIME-complete ([Y. Kazakov 2004](#));
 - $\text{Fin-Sat-}\mathcal{GC}^2$ is EXPTIME-complete ([IEP-H 2007](#)).